Xavier Maso

security, software, and cryptography engineer

he/him/his

(+33) 621-822-328 contact@xaviermaso.com www.xaviermaso.com

Skills

Security

Code and Design Reviews	Scripting	Reverse Engineering	Exploitation	Pentesting
★★★★★	★★★★★	★★★★★	★★★★★	★★★★★

Software development

Python / Ruby	Bash	C / Rust	ASM x86	Java
★★★★★	★★★★★	★★★★★	★★★★★	★★★★★

JavaScript / TypeScript	Nix	Elm-Lang	Haskell / OCaml	Typst
★★★★★	★★★★★	★★★★★	★★★★★	★★★★★

System administration

Linux: NixOS, Ubuntu, Debian, Fedora.
Configuration as Code, automation: NixOps, Docker, Ansible, GitHub Actions

Soft skills

Team management: Implement the environment which allows to develop the products and achieve goals.
Agile: Xtreme Programming proponent, used to Agile methodology, especially Scrum, and Lean approaches.
Speaker: Animation and presentation about a product, a project or to share experiences.
Interpersonnal: Listening and understanding individuals to facilitate interactions within a group.

Professional Experiences

DINUM *September 2026 – Today*
DevSecOps Engineer *Remote*

Part of the Direction Interministérielle du Numérique (DINUM), the OPI team builds a PaaS platform to host open-source public services managed by diverse French state ministries and agencies.

Laboratoire d'Innovation de l'ANSSI *July 2025 – August 2026*
Contractor, CISO *Remote*

The Laboratoire d'Innovation is a state startup incubator of the "Agence Nationale de la Sécurité des Systèmes d'Information", member of the BetaGouv ecosystem. It leverages product design and entrepreneurship methodologies to help improve the efficiency of the agency's actions. The team counts about twenty people, and a dozen developpers following Xtreme Programming practices.

- Manage the security of the information system and all the services (5-10) developped at the Lab'.
- Information system administration: operate infrastructure and systems necessary for the team and the products to work.
- Application Security and Development: provide expertise and technical support on security by design and secure coding practices, while contributing to all the products' development.
- Services procurement: define and scope needs, list and prioritize constraints, research commercial offers on the market, handle communications with third-parties, follow the process for public procurement.
- GRC: drive yearly homologation process, contribute to risk assessments, prioritize actions, follow external audits, write procedures.
- R&D: infrastructure as code to ease administation, SAST and DAST automation based of modern technologies to automatically identify and triage issues, use of securix based on NixOS for managed devices.

Omnicité

August 2025 – Today

Member *Remote*

Omnicité is a French cooperative of entrepreneurs, caring for a sustainable economy, and encouraging autonomy and mutual help.

PALO IT

March 2025 – July 2025

Tech Consultant, Software Engineer *Remote*

PALO IT France was the local branch of a global, B Corp certified, technology consultancy that develops and scales digital and sustainable products and services.

- Work on the technical aspect of several commercial proposals: understand needs, audit existing systems, survey technological solutions, draft change plans.
- Research and development for internal services: self-hosted VPN, consultants' resumes as code ease of maintenance for sales.

Oracle NetSuite

November 2021 – March 2025

Senior Application Security Engineer *Remote*

The Product Security Team assists product and engineering teams during the software development lifecycle of NetSuite.

- Collaborate on security design, implement security controls, and conduct code reviews.
- Build and manage tools (libraries, programs, processes, etc.) to help prevent and identify issues.
- Produce and deliver educational security material for engineers and product teams.

Tweag

May 2021 – September 2021

Open Source Fellow *Remote*

Tweag is a French software innovation lab running a fellowship program for those eager to contribute to open source projects.

- Work on fuzzing nix, the declarative package manager.
- Target definition and priorization; Write fuzzers; Explore using sanitizers; Run small experimental fuzzing sessions.

- Integrate to OSS-fuzz.

Arizona State University (ASU)
April 2019 – September 2021

Research Apprentice on Binary Analysis — *Phoenix, United States of America*

ASU's Security Engineering for Future Computing (SEFCOM) laboratory is a world-class cybersecurity research lab carrying out cutting-edge academic research.

- Design, implement, and evaluate a new, fast static binary analysis technique.
- Study state-of-the art: symbolic and concolic execution, fuzzing, automated exploit generation.
- Contribute to angr, the open-source binary analysis framework.
- Guest lecturer for CSE545 Software Security (Fall 2020): presentation, and live examples on using angr to implement a static binary analysis technique.
- Research on privacy.

Mozilla Corporation
May 2018 – August 2018

Web Security Intern, scanning modern web applications with OWASP ZAP — *London, United Kingdom*

The Firefox Operations Security team protects the core services and release engineering infrastructures Mozilla relies on to build, ship and run Firefox. OWASP ZAP is a free and open-sourced security tools to help automatically find security vulnerabilities in web applications.

- Research and development in Javascript and the DOM API: keep track of events and interactions happening in a webpage. Released the @zaproxy/front-end-tracker library on npm.
- Development of a ZAP add-on to create a framework for pentesters to run security scripts directly in the browser.
- Research on modern web applications (Single Page Applications) vulnerabilities.
- Part of a distributed team and contributing to an open-source community.

Orange Application for Business
June 2017 – August 2017

System Administrator, securing workstations — *Bordeaux, France*

OAB is a subsidiary of Orange S.A developing and hosting services for businesses. Integrated to an operations team of four, my mission was to secure their Linux workstations.

- System administration: Arch, Ubuntu, Debian.
- Define and apply good security practices for laptops and mobile devices.
- Development: ansible roles, audit-ssh-client-config script.
- Research: securing the boot process (anti-evil maid), NixOS, Public Key Infrastructures.
- Training and communication about the security practices: write documentation, create tutorials, run workshops.

RedShield Security Ltd
January 2015 – April 2016

Software Engineer — *Wellington, New Zealand*

Shielding-with-a-service company providing a web application firewall, vulnerability scanner and analyzer to assist IT departments ensure the security of their systems.

- Development in Ruby and Javascript; Debugging, and automate testing.· Configuration of production and CI servers (Jenkins).
- Security: offensive, secure development, incident response.

CER SNCF
June 2012 – October 2014

Full-stack Developer — *Bordeaux, France*

Distributed all over the Aquitaine and Poitou Charentes French regions, the CER (for Committee of Regionnal Establishment) provides services to SNCF’s workers.

- Web development, information system administration and maintenance, automation.
- IT service management: needs analysis; goals, processes, and tasks definition; listing of skills and resources to prioritize our work.
- Help in recruitment for IT service: define required skills, and mission scopes; Interview candidates. Project, subject of my master thesis
- Build a website dedicated to SNCF's employees: tasks and planning definition, resources and skills affectation to provide this new service within six months with a three person team.
- Communication and product launching: define and reach targets, follow the adoption and usage of the product.
- Integration within the existing information system: shared databases services, web server deployment.
- Server on Ubuntu 14.04, database is PostgreSQL, Nginx and Unicorn as web stack.
- Application built with Ruby-On-Rails, Slim, SASS and CoffeeScript. RSpec and Capybara for testing and QA.

Publications

[1] W. Gibbs *et al.*, "Operation Mango: Scalable Discovery of Taint-Style Vulnerabilities in Binary Firmware Services," in *33rd USENIX Security Symposium (USENIX Security 24)*, Philadelphia, PA: USENIX Association, Aug. 2024, pp. 7123–7139. [Online]. Available: https://www.usenix.org/conference/usenixsecurity24/presentation/gibbs

[2] S. Karami *et al.*, "Unleash the Simulacrum: Shifting Browser Realities for Robust Extension-Fingerprinting Prevention," in *31st USENIX Security Symposium (USENIX Security 22)*, Boston, MA: USENIX Association, Aug. 2022, pp. 735–752. [Online]. Available: https://www.usenix.org/conference/usenixsecurity22/presentation/karami

Education

University of Bordeaux *2016 – 2019*
Master of Mathematics and Applications: Cryptology and IT Security. *Bordeaux, France*

Ingésup *2009 – 2014*
Master of computer information technologies, specialty in Development *Bordeaux, France*

- Member of the security lab: research and learning about exploitation and penetration testing.
- Member of the games lab: participation in video game development contests.

Lycée La Boétie *2005 – 2009*
French Baccalauréat in Sciences, specialty in Mathematics; Obtained with honors. *Bordeaux, France*

Volunteer

Les Petites Cantines Bordeaux *May 2026 – Today*
Co-Administrator *Bordeaux, France*

A participative, at free price, and non-profit restaurant: patreons can contribute to the making of the meal, pay what they want, all under a volunteer-based governance.

Okiwi *June 2023 – Today*
Secretary *Bordeaux, France*

Okiwi is an open community of software craftsmanship practicioners, from Bordeaux. It advocates for benevolence, inclusion, sharing, openness, and diversity.

Proxité
June 2024 – Today
Mentor *Bordeaux, France*

Proxité pairs volunteers with young people from socioeconomically disadvantaged neighbourhoods, to help them stick with school, support sustainable profesionnal insertion, and strengthen local social circles.

Nix/NixOS community
2018 – Today
Contributor *Remote*

Contribute to Nixpkgs, Nix, NixOS, other related projects, and the community forums.

DjangoGirls Bordeaux
2016 – 2019
Volunteer *Bordeaux, France*

Non-profit community that empowers and helps women get involved with programming. Organization and coaching for DjangoGirls events, and recurring meetups. Teaching programming basics using Python.

Hobbies

Challenges	Reading	Sports
root-me, hackthebox, exercism	sciences, economy, politics, psychology, sociology	muay thai, chess, hiking, diving (PADI AOW), former judoka, football, and handball player